Document Extraction Engine: Product Requirements Document (PRD)

Oct 28, 2026 · @Praveen

This PRD defines an on-premise, schema-driven engine that turns scanned BFSI and healthcare documents into validated, evidence-grounded JSON, written so each requirement can be built and tested as a spec. It merges the Technical Approach document (architecture, models, 7 document types) with the Product Requirements Outline (personas, F1–F21, roadmap), and adds mock-screen, guardrail, security and NIST CSF 2.0 requirements.

1. Decisions taken in this PRD

The two inputs disagree in five places. This PRD resolves each one so engineering is not blocked; each decision is listed again in Section 15 for sign-off.

#	Conflict	Approach document	Outline	PRD decision
D1	Deployment	100% on-premise, zero data leaves premises	SaaS first (open-question answer)	**Single-tenant, customer-controlled deployment first** (customer VPC or on-prem, same container build). A vendor-hosted multi-tenant SaaS is deferred until a regulator-approved model exists for PII/PHI.

#	Conflict	Approach document	Outline	PRD decision
D2	Model providers	Local models only; cloud APIs excluded	Provider abstraction, swap by cost or residency	**Provider abstraction is kept, but only local providers are allowed in the default build.** External API providers are disabled by a deployment policy flag and fail closed.
D3	Accuracy bar	Field F1 > 92% (structured/semi), CER < 3%, WER < 6%	95%+ key-field accuracy; open answer "80% above" for STP	**Two bars:** F1 ⩾ 92% on all fields and ⩾ 95% on schema-marked key fields for structured and semi-structured types; STP ⩾ 60% at MVP. Handwritten prescriptions are exempt from STP and always route to review below the confidence threshold.
D4	Latency	< 3.0 s/page on GPU	< 10 s/page	**p95 ⩽ 3.0 s/page on the reference GPU; ⩽ 10 s/page on CPU-only fallback.**
D5	MVP scope	7 types, but prototype on CASA + Discharge Summary first	2–3 types across InsurTech, FinTech, Healthcare	**MVP = CASA Account Opening, Insurance Application, Discharge Summary** (one per domain). The other four types are schema-only additions in v1.

Two further inputs from the outline's answers are adopted as stated: pricing is per page, and launch languages are English, Hindi, Telugu, Tamil and Malayalam. Malayalam is not covered by the approach document's model selection, so it is a validated-or-not gate in Section 14 rather than an assumption.

2. Product context

Banks, insurers and hospitals receive proposal forms, claim forms, loan and card applications, discharge summaries and prescriptions in hundreds of layouts. Template OCR breaks on each new layout, flat OCR destroys tables and checkbox state, and manual keying is slow and error-prone. Cloud OCR and LLM APIs are not an option because PII and PHI cannot leave the premises.

Vision. Onboard a new document type by writing a schema, not by building templates or retraining. Every value is typed, validated, confidence-scored and linked to the pixel region it came from. Anything the engine is unsure about goes to a person instead of being guessed.

2.1 Personas

Persona	Primary need	Key screens
Operations analyst	Upload, track and export documents with minimal rework	SCR-01, SCR-02, SCR-06
Reviewer / QC	See only exceptions, verify against the source in one click, correct inline	SCR-03, SCR-04
Business / product owner	Add or change a document type without engineering	SCR-05
Developer / integrator	Stable API, webhooks, versioned output contract	SCR-07
Compliance and audit	Traceability, access control, retention evidence	SCR-08, SCR-09
Platform / security admin	Deploy on-prem, manage users, models, keys and egress policy	SCR-09, SCR-10

2.2 Goals

1. Extract the 7 launch document types into strictly typed JSON with 100% syntactically valid output.

2. Make every value traceable to a page and bounding box, with a calibrated confidence.
3. Keep 100% of document data inside the customer's boundary, including inference and fine-tuning.
4. Let a business user add a new document type in under one day, with no code change.
5. Route uncertain fields to human review instead of guessing, especially handwriting.

2.3 Non-goals (this release)

- Underwriting, credit or claim decisions, or any automated action on extracted data.
- Document generation, long-term archival or document management.
- Cloud OCR/LLM API calls, and vendor-hosted multi-tenant SaaS (see D1, D2).
- Free-form question answering over documents.
- Training a foundation model; only parameter-efficient fine-tuning (PEFT/QLoRA) on customer data, run locally.

2.4 Success metrics

Targets are measured on a labelled benchmark set per document type (Section 13) and gate each phase (Section 14).

ID	Metric	Definition	Target
M1	Field F1	Harmonic mean of precision and recall on key-value pairs, structured and semi-structured types	⩾ 92% all fields; ⩾ 95% key fields
M2	Parser CER / WER	Levenshtein error of layout-parser text vs ground truth, printed text	CER < 3%, WER < 6%
M3	Schema adherence	Outputs passing schema validation with zero exceptions	100%
M4	Classification accuracy	Documents routed to the correct type; unknown routed to review	⩾ 98%
M5	Straight-through rate (STP)	Documents with no human touch	⩾ 60% at MVP; ⩾ 80% at v1 on structured types

ID	Metric	Definition	Target
M6	Confidence calibration	Gap between stated confidence and measured accuracy	⩽ 3 points
M7	Latency	Upload to validated JSON, per page	p95 ⩽ 3.0 s GPU; ⩽ 10 s CPU fallback
M8	New document type onboarding	Schema defined to production-approved	< 1 working day
M9	Data egress	Outbound connections carrying document data	0
M10	Review efficiency	Median reviewer time per flagged field	⩽ 15 s (to validate)

3. Scope and how to read this spec

3.1 Launch document types

Seven types in two domains. Complexity follows the approach document's matrix: structured forms are layout-predictable, handwriting is not.

Doc ID	Document	Domain	Category	Phase	Main technical challenge
DOC-01	CASA Account Opening Form	BFSI	Structured	MVP	Character-grid fields, checkbox marks, KYC blocks
DOC-02	Insurance Application Form	BFSI	Structured	MVP	Medical questionnaires, dense tick grids, nominees
DOC-03	Hospital Discharge Summary	Healthcare	Semi-structured	MVP	Hospital-specific layouts, multi-page medicine tables
DOC-04	Loan Application Form	BFSI	Structured	v1	Multi-column financial tables, applicant vs co-applicant
DOC-05	Credit Card Application	BFSI	Structured	v1	Disclosure tables, `[x]` vs `[ ]` disambiguation
DOC-06	Insurance Claim Form	BFSI	Semi-structured	v1	Itemised bills must sum to the claimed total
DOC-07	Doctor's Prescription	Healthcare	Unstructured	v1	Cursive handwriting, shorthand (OD, BD, TDS, SOS), ambiguous drug names

Product brochures and policy wordings from the outline are tracked as DOC-08 (v1, fact extraction from unstructured text) and are not in the launch seven.

3.2 Inputs, languages and output

- **Input formats:** PDF (native and scanned, multi-page), JPG, PNG, TIFF at MVP; DOCX, XLSX, email and HTML in v1.
- **Languages at launch:** English, Hindi, Telugu, Tamil, Malayalam. Printed text first; handwriting in English only at MVP.
- **Output:** JSON (canonical), plus CSV and XML renderings in v1. Dates ISO 8601, amounts as decimals with ISO 4217 currency, booleans as true/false.

3.3 Spec conventions (for spec-driven development)

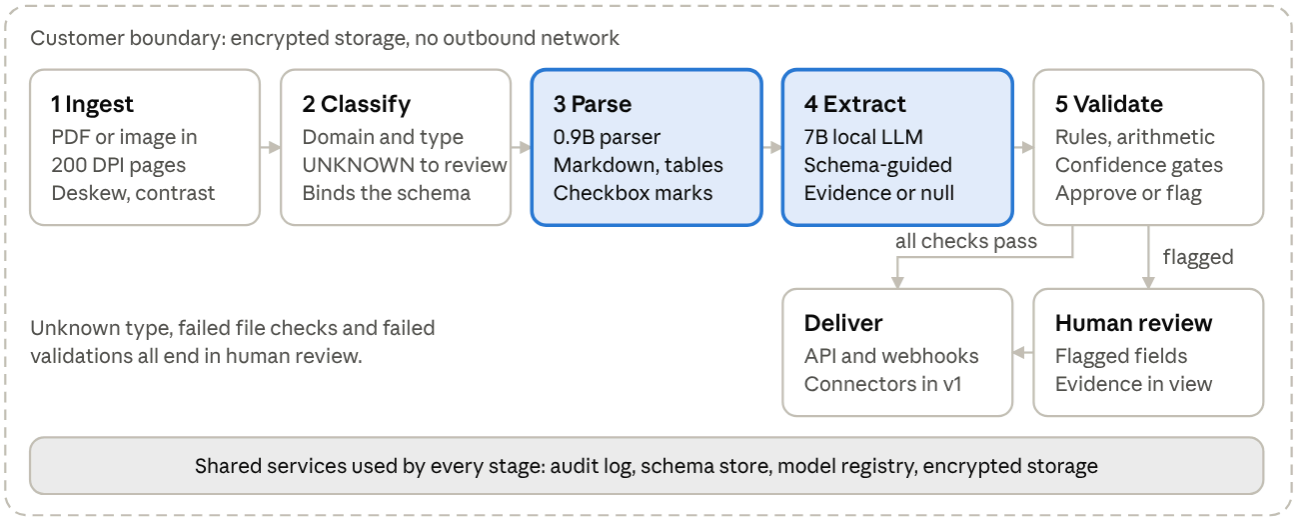
This document is the source of truth that code, tests and mock screens are generated from. To make that reliable:

- **Requirement IDs** are stable: `FR-<AREA>-nnn`, `SCH-nnn`, `API-nnn`, `SCR-nn`, `MOCK-nnn`, `GR-nnn`, `SEC-nnn`, `NFR-nnn`. Never reuse an ID; retire it instead.
- **Wording** follows EARS: *The system SHALL …* (ubiquitous), *WHEN … the system SHALL …* (event), *IF … THEN the system SHALL …* (unwanted case). SHALL is mandatory, SHOULD is recommended, MAY is optional.
- **Priority:** P0 = MVP gate, P1 = v1, P2 = later. Each row carries its phase through priority.
- **Acceptance criteria** are written so they can become a test without interpretation: Given / When / Then, or a measurable threshold with the benchmark that measures it.
- **Traceability:** every requirement cites its source (approach document section, or outline F-number) so that a changed source flags the requirements to re-check.
- **Ambiguity rule:** if a requirement cannot be turned into a test, it is a defect in this PRD. Raise it as a comment; do not guess in code.

4. Architecture

The engine is a two-stage decoupled pipeline: a small vision model reads the page layout, then a local LLM extracts schema fields from that text, so each can change independently.

Two decoupled models inside one customer boundary



target architecture · 5 pipeline stages, 1 review loop

Stages 3 and 4 (highlighted) are separate models, so either can be upgraded or re-quantised without touching the other. Only fully validated documents are approved automatically; everything else goes to a person.

ID	Architectural requirement	Pri
ARC-001	All components SHALL run inside the customer boundary and SHALL NOT call any external service (see GR-013, SEC-011).	P0
ARC-002	The parser and the extractor SHALL be separate services behind versioned interfaces so that either can be replaced by configuration (NFR-008).	P0
ARC-003	Stages SHALL communicate through a durable queue; workers SHALL be stateless, and each stage SHALL store its output under the run ID.	P0
ARC-004	Model serving SHALL run in network-isolated containers with access only to the stage's input and output (GR-007).	P0
ARC-005	The reference stack from the approach document (Python, pypdfium2, OpenCV, Pydantic, instructor or Outlines, Ollama or vLLM, PaddleOCR-VL-1.6 or GLM-OCR, quantised Qwen2.5-7B-Instruct) is the starting point and SHALL be replaceable if the interfaces and measured outcomes hold.	P0

5. Functional requirements

Organised in pipeline order. Source column: `A` = Technical Approach document section, `F` = Outline requirement number.

5.1 Ingestion and preprocessing

ID	Requirement	Pri	Acceptance criteria	Source
FR-ING-001	The system SHALL accept PDF, JPG, PNG and TIFF via API, UI upload and a watched folder.	P0	Given each format, When uploaded, Then a document record is created and status becomes `queued` within 2 s.	F1
FR-ING-002	The system SHALL verify each file by magic bytes, size limit and page limit, and reject others with an error code.	P0	Given a renamed `.exe` as `.pdf`, Then upload is rejected `UNSUPPORTED_FILE` and nothing is stored.	GR-001
FR-ING-003	The system SHALL scan each file for malware before processing.	P0	Given the EICAR test file, Then status is `rejected_malware` and no pipeline stage runs.	SEC-009
FR-ING-004	The system SHALL split multi-page PDFs and render each page at 200 DPI.	P0	Given a 10-page PDF, Then 10 page images at 200 DPI exist, in order, linked to the document.	A §3
FR-ING-005	The system SHALL de-skew, correct rotation and normalise contrast per page, and record the transforms applied.	P0	Given a benchmark page skewed 5°, Then residual skew ⩽ 0.5° and the transform is in the page record.	A §3, F2
FR-ING-006	The system SHALL treat a repeated upload of the same file hash within one tenant as idempotent when the client sends the same idempotency key.	P0	Given two POSTs with one key, Then one document exists and both calls return its ID.	NFR-006
FR-ING-007	The system SHOULD split a bundled file into logical documents and group related pages.	P1	Given a 12-page bundle of 3 documents, Then 3 child documents with correct page ranges, ⩾ 95% boundary accuracy on the benchmark.	F3

ID	Requirement	Pri	Acceptance criteria	Source
FR-ING-008	The system SHOULD accept DOCX, XLSX, email and HTML by converting to page images or text with layout.	P1	Each format yields the same output contract as a scanned PDF.	F1

5.2 Classification and routing

ID	Requirement	Pri	Acceptance criteria	Source
FR-CLS-001	The system SHALL classify each document into a domain (BFSI or Healthcare) and a document type, returning a confidence in [0, 1].	P0	Classification accuracy ⩾ 98% (M4) on the benchmark.	A §3 stage 2, F5
FR-CLS-002	IF classification confidence is below the tenant threshold OR no schema matches, THEN the system SHALL set type `UNKNOWN` and route the document to review; it SHALL NOT run extraction.	P0	Given an unrelated utility bill, Then type is `UNKNOWN`, no extraction output exists, a review task is created.	F5, GR-004
FR-CLS-003	WHEN a type is resolved, the system SHALL bind the current approved version of its schema and record that version on the document.	P0	Output envelope contains `schema_id` and `schema_version` matching the approved version at processing time.	A §3 stage 2
FR-CLS-004	The reviewer SHALL be able to override the document type, which re-runs extraction with the chosen schema.	P0	Given an override, Then a new extraction run is created and the old one is retained in history.	F16

5.3 Visual layout parsing

ID	Requirement	Pri	Acceptance criteria	Source
FR-PRS-001	The system SHALL convert each page to layout-faithful Markdown with HTML tables, using the configured local parser model, and keep text blocks with page and bounding box.	P0	Given a benchmark page, Then output has per-block `bbox` and CER < 3%, WER < 6% on printed text (M2).	A §3 stage 3, F6
FR-PRS-002	The system SHALL render every checkbox and tick as `[x]` or `[ ]` and record the mark's bounding box.	P0	Checkbox state accuracy ≥ 98% on the DOC-02 and DOC-05 benchmark sets.	A §4.1, F6
FR-PRS-003	The system SHALL merge segmented character grids (for example `|J|O|H|N|`) into one value before extraction.	P0	Given a PAN written in 10 boxes, Then the parsed value has no spaces and equals ground truth.	A §4.1
FR-PRS-004	The system SHALL detect tables, key-value pairs, signatures and stamps as typed layout elements.	P0	Element detection recall ≥ 90% on the benchmark.	F6
FR-PRS-005	The system SHALL report an optical confidence per text block.	P0	Every block has `ocr_confidence` in [0, 1].	A §4.2
FR-PRS-006	The system SHOULD read English handwriting and flag struck-out or overwritten entries.	P1	Handwriting fields below the threshold are `review_required`; none is auto-approved.	F7
FR-PRS-007	The system SHOULD read Hindi, Telugu, Tamil and Malayalam printed text.	P1	Per-language CER reported on the benchmark; a language ships only if its CER meets M2.	F4

5.4 Extraction

ID	Requirement	Pri	Acceptance criteria	Source
FR-EXT-001	The system SHALL extract exactly the fields defined in the bound schema from the parsed text, using the local LLM, with no per-layout template or retraining.	P0	A document of a known type from an unseen layout meets M1 without code changes.	A §3 stage 4, F8
FR-EXT-002	The system SHALL constrain model output to the schema using grammar-guided decoding.	P0	Over 1,000 runs, zero schema-validation exceptions (M3).	A §3 stage 4
FR-EXT-003	IF a field is absent or not legible, THEN the system SHALL return `null` with `null_reason` of `absent` or `illegible`; it SHALL NOT infer a value.	P0	Given a form with the PAN left blank, Then `pan_card_number` is null, reason `absent`.	A §7.2, F11
FR-EXT-004	The system SHALL return, for every non-null value, its source `page`, `bbox` and the supporting text snippet.	P0	100% of non-null values have evidence; evidence text contains the value after normalisation.	F15, GR-002
FR-EXT-005	The system SHALL extract repeating groups (nominees, dependents, medicines, line items) as arrays with stable order and, where present, a parent link.	P0	Given 3 nominees, Then 3 array items in page order, each with evidence.	F9
FR-EXT-006	The system SHALL associate figures with the right party (applicant or co-applicant) in multi-column tables.	P1	Party-attribution accuracy ≥ 95% on DOC-04.	A §2
FR-EXT-007	The system SHALL map checkbox markers to booleans or enumerated values defined in the schema.	P0	`[x]` yields true, `[ ]` yields false, no mark yields null.	A §4.1

ID	Requirement	Pri	Acceptance criteria	Source
FR-EXT-008	The system MAY run a second reconciliation pass on fields below the confidence threshold and keep the result only if it agrees with evidence.	P1	Second pass raises key-field F1 on the benchmark without raising hallucination rate.	F7, outline §7

5.5 Normalisation and validation

ID	Requirement	Pri	Acceptance criteria	Source
FR-VAL-001	The system SHALL normalise dates to ISO 8601, amounts to decimals with currency, phone to E.164, and map values to tenant code lists.	P0	`22/06/1985` becomes `1985-06-22`; `+91 98765 43210` becomes `+919876543210`.	F12
FR-VAL-002	The system SHALL validate format rules (PAN, Aadhaar checksum, IFSC, policy numbers) and mark failures `validation_failed`.	P0	Given `ABCDE1234` as PAN, Then flagged `validation_failed` and routed to review.	A §3 stage 5, F13
FR-VAL-003	The system SHALL verify arithmetic: the sum of itemised amounts equals the declared total, within a configured tolerance.	P0	Given items summing to 41,500 and a total of 45,100, Then an `ARITHMETIC_MISMATCH` flag is raised on the claim.	A §4.1
FR-VAL-004	The system SHALL enforce chronological and cross-field rules (for example discharge date ⩾ admission date).	P0	Given discharge before admission, Then both fields are flagged and the document is routed to review.	A §4.2, F13
FR-VAL-005	The system SHALL map clinical frequency shorthand (OD, BD, TDS, SOS) to standard descriptions and keep the original text.	P1	`BD` yields `Twice daily (BD)`; unknown abbreviations are kept raw and flagged.	A §4.2

ID	Requirement	Pri	Acceptance criteria	Source
FR-VAL-006	Rules SHALL be configurable per schema and versioned with it, without a code release.	P0	Changing a rule creates a new schema version; old documents keep the version they ran with.	outline §5
FR-VAL-007	The system SHOULD check consistency across documents in one case (name, date of birth, address).	P2	Mismatch across KYC and application raises a case-level flag.	F13, F20

5.6 Confidence, routing and review

ID	Requirement	Pri	Acceptance criteria	Source
FR-CNF-001	The system SHALL compute a calibrated confidence per field from parser confidence, model signals and validation outcome.	P0	Calibration gap ⩽ 3 points (M6) on the benchmark.	F14
FR-CNF-002	The system SHALL set `review_required: true` on any field whose confidence is below its threshold; default threshold 0.75 for handwriting-derived fields, tenant-configurable per field.	P0	A handwritten drug name at 0.62 is `review_required` and not auto-approved.	A §4.2, F14
FR-CNF-003	WHEN all fields pass validation and thresholds, the system SHALL auto-approve the document; otherwise it SHALL create a review task listing only the flagged fields.	P0	Review task contains flagged fields only; STP counted from auto-approved documents (M5).	F16
FR-REV-001	The reviewer UI SHALL show the page image beside the extracted fields and highlight the evidence box for the selected field.	P0	Selecting a field scrolls and highlights its bbox within 300 ms.	F15, F16

ID	Requirement	Pri	Acceptance criteria	Source
FR-REV-002	The reviewer SHALL be able to accept, correct or mark-as-null a field inline; each action SHALL be audit-logged with user, time, old and new value.	P0	Audit log contains one entry per action.	F16
FR-REV-003	A corrected value SHALL be re-validated before the task can be completed.	P0	An invalid correction blocks completion with the failing rule shown.	F13
FR-REV-004	The system SHOULD add reviewer corrections to the benchmark candidate set, subject to approval.	P1	Corrections appear in a pending-ground-truth list; none enters the benchmark automatically.	F17, GR-012

5.7 Delivery and audit

ID	Requirement	Pri	Acceptance criteria	Source
FR-OUT-001	The system SHALL return results by synchronous API (single page) and asynchronous API with polling or webhook.	P0	Contract tests in Section 7 pass.	F18
FR-OUT-002	The system SHALL emit output in the canonical JSON envelope defined in Section 6.	P0	Output validates against the published JSON Schema.	F18
FR-OUT-003	The system SHOULD offer CSV and XML renderings and prebuilt connectors to a policy-admin, loan-origination, claims or CRM system.	P1	One connector passes an end-to-end test against a customer sandbox.	F19
FR-AUD-001	The system SHALL record every extraction run with input hash, model versions, prompt version, schema version, timings and outputs so that it can be reproduced.	P0	Re-running a stored run in the same environment yields identical output (deterministic decoding).	outline NFR

6. Schema and output contract

The schema is the main control surface: business users define what to extract and which rules apply, and the engine adapts. Each schema compiles to a typed model (Pydantic) and a JSON Schema used for grammar-guided decoding and for output validation.

6.1 Schema requirements

ID	Requirement	Pri	Acceptance criteria
SCH-001	A schema SHALL define per field: name, type (string, integer, decimal, boolean, date, enum, object, array), description, required or optional, allowed values, format or regex, evidence required (default true), confidence threshold, and PII class.	P0	A schema missing `type` for any field fails publish validation.
SCH-002	Schemas SHALL support nested objects and repeating groups.	P0	A schema with `nominees[]` produces an array in the output.
SCH-003	A schema SHALL be authored as YAML or JSON and through the visual editor (SCR-05); both SHALL round-trip without loss.	P0	Export from editor, re-import, diff is empty.
SCH-004	A schema SHALL move through states `draft`, `testing`, `approved`, `retired`; only `approved` schemas are used in production.	P0	Attempt to bind a `draft` schema in production returns `SCHEMA_NOT_APPROVED`.
SCH-005	Promotion to `approved` SHALL require a passing test-harness run and approval by a second user (four-eyes).	P0	Same user cannot approve own schema; harness result is attached to the approval.
SCH-006	Every schema change SHALL create an immutable version with author, time and diff; rollback SHALL re-activate a prior version without deleting history.	P0	After rollback, new documents bind the older version; the audit log shows the rollback.
SCH-007	The test harness SHALL run a schema on uploaded sample documents with ground truth and report precision, recall, F1 and calibration per field.	P0	Harness report lists every field with its F1.

ID	Requirement	Pri	Acceptance criteria
SCH-008	Reference data (code lists, lookup tables) SHALL be uploadable and versioned per tenant.	P1	A state-code list is referenced by field `state` and applied in normalisation.
SCH-009	A schema library of cloneable prebuilt schemas SHALL ship for the seven launch types.	P1	Each launch type has an approved library schema.
SCH-010	The system MAY draft a schema from a plain-English description, always as `draft` for human editing.	P2	Generated schema is never auto-approved.

6.2 Schema definition format (illustrative)

```
schema_id: casa_account_opening
version: 3
status: approved
domain: BFSI
fields:
  - name: applicant.full_name
    type: string
    required: true
    description: Applicant name, merged from character boxes, uppercase as
printed
    pii_class: direct_identifier
    confidence_threshold: 0.90
  - name: applicant.pan_card_number
    type: string
    required: false
    format: '^[A-Z]{5}[0-9]{4}[A-Z]$'
    pii_class: government_id
  - name: account_details.initial_deposit_amount
    type: decimal
    currency: INR
  - name: account_details.opt_in_services.cheque_book
    type: boolean
    source: checkbox
rules:
  - id: R1
    expression: applicant.date_of_birth <= today - 18y
    on_fail: review
```

6.3 Output envelope

Every value is wrapped with its confidence and evidence. The envelope below is the P0 contract; field names are normative.

```
{
  "document_id": "d_01HZX...",
  "document_metadata": {
    "document_type": "CASA_ACCOUNT_OPENING",
    "schema_id": "casa_account_opening",
    "schema_version": 3,
    "classification_confidence": 0.96,
    "status": "needs_review",
    "page_count": 2
  },
  "fields": {
    "applicant.full_name": {
      "value": "SURESH KUMAR RAMAN",
      "confidence": 0.97,
      "null_reason": null,
      "review_required": false,
      "validation": "passed",
      "evidence": { "page": 1, "bbox": [112, 240, 610, 281], "text": "S U R E
S H ..." }
    },
    "applicant.pan_card_number": {
      "value": null,
      "confidence": 0.0,
      "null_reason": "absent",
      "review_required": false,
      "validation": "not_applicable",
      "evidence": null
    }
  },
  "flags": [ { "code": "ARITHMETIC_MISMATCH", "fields": ["claim.total"] } ],
  "run": { "run_id": "r_...", "model_versions": { "parser": "...",
"extractor": "..." }, "prompt_version": "..." }
}
```

ID	Requirement	Pri
SCH-011	The envelope SHALL be published as JSON Schema and versioned with semantic versioning; breaking changes SHALL bump the major version and keep the previous version available for at least two releases.	P0
SCH-012	`status` SHALL be one of `processing`, `auto_approved`, `needs_review`, `reviewed`, `rejected`, `failed`.	P0
SCH-013	`bbox` SHALL be `[x0, y0, x1, y1]` in pixels of the 200 DPI page image with origin top-left.	P0
SCH-014	Fields marked PII in the schema SHALL be maskable in API responses by caller permission (see SEC-007).	P0

7. API and integration specification

REST over HTTPS inside the customer boundary, JSON bodies, OpenAPI 3.1 published from the code. All calls need a bearer token with scopes (SEC-002).

ID	Method and path	Purpose	Pri
API-001	`POST /v1/documents`	Upload a file (multipart) with optional `document_type_hint`, `schema_id`, `callback_url`, `Idempotency-Key`; returns `202` with `document_id`.	P0
API-002	`GET /v1/documents/{id}`	Status, metadata and, when ready, the output envelope.	P0
API-003	`GET /v1/documents/{id}/pages/{n}`	Page image (for the review UI), subject to permission.	P0
API-004	`POST /v1/documents/{id}/extract-sync`	Single-page synchronous extraction, timeout 30 s.	P0
API-005	`GET /v1/review-tasks` and `POST /v1/review-tasks/{id}/complete`	List and complete exception tasks with corrections.	P0

ID	Method and path	Purpose	Pri
API-006	`GET/POST /v1/schemas`, `POST /v1/schemas/{id}/versions`, `POST /v1/schemas/{id}/approve`	Schema lifecycle.	P0
API-007	`POST /v1/schemas/{id}/test-runs`	Run the harness on a sample set.	P0
API-008	`POST /v1/webhooks`, `GET /v1/audit-events`	Subscriptions and audit export.	P0 / P1
API-009	`GET /v1/health`, `GET /v1/metrics`	Liveness, readiness and Prometheus metrics.	P0

7.1 Events

Webhooks are signed (HMAC-SHA256 with a per-subscription secret), retried with exponential back-off for 24 hours, and delivered at least once, so consumers de-duplicate by `event_id`.

Event	Fired when
`document.completed`	Status becomes `auto_approved` or `reviewed`
`document.needs_review`	A review task is created
`document.failed`	A stage fails after retries
`schema.approved`	A schema version is approved

7.2 Error model

ID	Requirement	Pri
API-010	Errors SHALL use one shape: `{ "error": { "code", "message", "request_id", "details" } }`, with stable machine-readable codes (`UNSUPPORTED_FILE`, `SCHEMA_NOT_APPROVED`, `DOCUMENT_NOT_FOUND`, `RATE_LIMITED`, `VALIDATION_FAILED`).	P0
API-011	Error messages SHALL NOT contain extracted document content or PII.	P0

ID	Requirement	Pri
API-012	`429` responses SHALL carry `Retry-After`; clients exceeding the tenant rate limit SHALL be throttled, not dropped.	P0
API-013	Mutating calls SHALL accept `Idempotency-Key`; replays within 24 hours return the original result.	P0
API-014	Contract tests generated from the OpenAPI document SHALL run in CI and block release on any breaking change without a major version bump.	P0

8. Screens and mock-screen generation

Mock screens are generated from this section before any UI code is written, so every screen is specified, reviewed with the personas and traceable to a requirement. A screen without a spec here SHALL NOT be built.

8.1 Mock-screen generation requirements

ID	Requirement	Pri	Acceptance criteria
MOCK-001	Each screen SHALL have a screen spec containing: purpose, persona, entry points, layout regions, components, data bindings (API field paths), states, actions and transitions, validation messages, accessibility notes and linked requirement IDs.	P0	All ten screens in 8.2 have every listed element; a lint script fails the build on a missing element.
MOCK-002	Mocks SHALL be generated for all ten screens at desktop (1440 px) and tablet (1024 px) widths, in light and dark themes. SCR-03 is desktop-first.	P0	10 screens × 2 widths × 2 themes rendered and stored with the spec version.

ID	Requirement	Pri	Acceptance criteria
MOCK-003	Mocks SHALL render every state in the screen's state matrix (empty, loading, populated, partial failure, error, permission denied, and screen-specific states), not only the populated state.	P0	State matrix coverage report shows 100% of listed states rendered.
MOCK-004	Mock data SHALL be synthetic. No real customer document, name, ID number or health data SHALL be used in any mock, prompt or screenshot.	P0	Data-source manifest for mocks lists only generated or fictional data; a PII scanner on the mock repository passes.
MOCK-005	The system SHALL provide a synthetic document generator that produces scanned-looking samples (skew, noise, stamps, checkboxes, handwriting-style fonts) for all seven launch types, with matching ground-truth JSON, used for mocks, demos and tests.	P0	Each launch type has ⩾ 20 synthetic samples with ground truth; none derived from a real document.
MOCK-006	Every interactive element in a mock SHALL be annotated with its requirement ID and data binding so that developers and test authors can trace it.	P0	Annotation coverage ⩾ 95% of interactive elements.
MOCK-007	Mocks SHALL be linked into clickable flows for the primary journeys in 8.3.	P0	Each journey can be walked from its entry screen to its end state.
MOCK-008	Confidence and status SHALL be conveyed by text or icon as well as colour; contrast SHALL meet WCAG 2.2 AA; Indic scripts (Devanagari, Telugu, Tamil, Malayalam) SHALL render correctly in the field viewer.	P0	Accessibility checklist passes; a rendered sample in each script has no tofu characters.

ID	Requirement	Pri	Acceptance criteria
MOCK-009	Mocks SHALL be reviewed by one representative per persona and signed off; the sign-off, version and comments SHALL be recorded. A change to a mock that alters behaviour SHALL first change this section.	P0	Sign-off record exists for each screen before its build task starts.
MOCK-010	Mock generation SHALL be repeatable from a prompt template (8.5) plus the screen spec, so a regenerated mock after a spec change differs only where the spec changed.	P1	Diff of two generations from an unchanged spec is limited to non-semantic differences, reviewed by a person.

8.2 Screen inventory

Screen	Name	Persona	Purpose	Key components	Pri
SCR-01	Upload and intake	Ops analyst	Submit files or batches and see them accepted	Drop zone, type hint selector, batch list, validation errors	P0
SCR-02	Document list and status	Ops analyst	Find and track documents	Filterable table (status, type, date, confidence), bulk actions, progress	P0
SCR-03	Review workbench	Reviewer	Verify flagged fields against the source	Page viewer with highlight, field panel, confidence chips, flags, actions	P0
SCR-04	Review queue	Reviewer	Pick the next exception	Queue table, SLA age, assignee, claim/release	P0
SCR-05	Schema studio	Business owner	Define, test and approve schemas	Field tree editor, YAML view, rules editor, test harness results, version history	P0
SCR-06	Results and export	Ops analyst	View and export approved output	Envelope viewer, JSON/CSV/XML export, evidence drill-down	P0
SCR-07	Developer console	Integrator	Manage API keys, webhooks, view logs	Key list, webhook subscriptions, delivery log, API reference link	P0
SCR-08	Audit log explorer	Compliance	Reconstruct who did what	Event table, filters, run reproduction view, export	P0

Screen	Name	Persona	Purpose	Key components	Pri
SCR-09	Security and policy settings	Admin, compliance	Roles, retention, egress policy, masking rules	Role matrix, retention rules, provider policy, key status	P0
SCR-10	Operations dashboard	Admin, product owner	Watch accuracy, STP, latency, drift, cost	KPI tiles, confidence histogram, error patterns, model versions	P1

8.3 Primary journeys to prototype

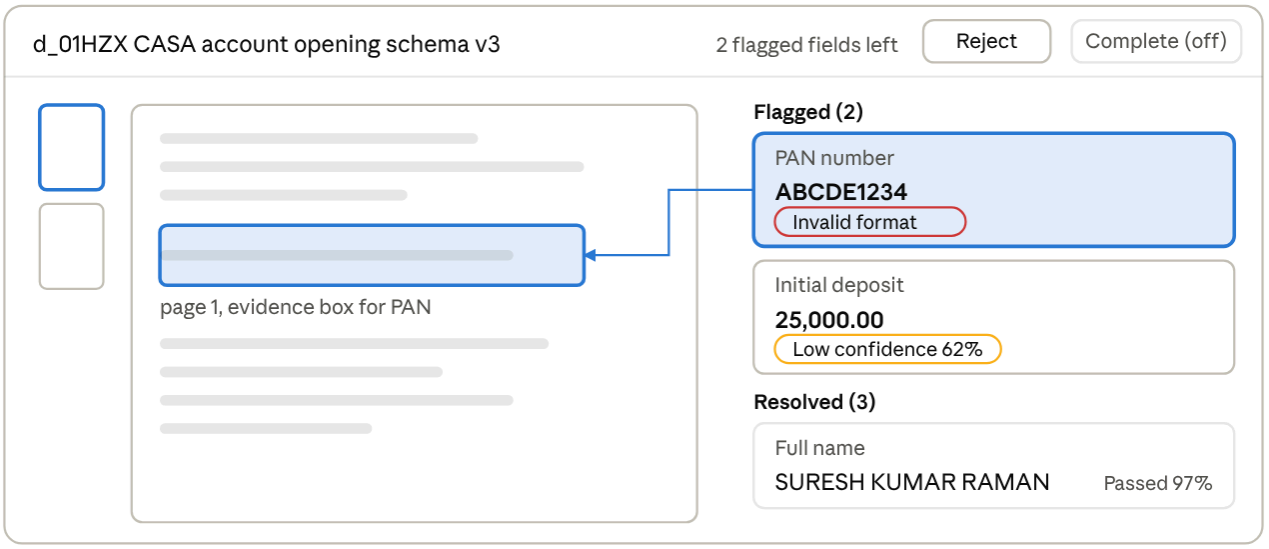
1. **Happy path.** SCR-01 upload → SCR-02 status moves to `auto_approved` → SCR-06 view and export.
2. **Exception path.** SCR-02 shows `needs_review` → SCR-04 claim task → SCR-03 correct two fields → complete → SCR-06.
3. **New document type.** SCR-05 define schema → run test harness → request approval → second user approves → SCR-01 upload a sample of the new type.
4. **Audit.** SCR-08 search a document → open run → see model, prompt and schema versions and every correction.

8.4 Screen spec: SCR-03 Review workbench (reference example)

Every other screen follows this level of detail.

- **Layout.** Left 55%: page viewer with zoom, pan and page thumbnails. Right 45%: field panel grouped by schema section, flagged fields first. Top bar: document ID, type, schema version, overall status, Complete and Reject buttons.
- **Field row.** Label, value (editable), confidence chip (text plus icon), validation state, null reason, evidence snippet. Selecting a row highlights its bounding box in the viewer; clicking a box selects its row.
- **Actions.** Accept, Edit, Mark as null, Change document type, Add comment, Next flagged field (keyboard `N`), Complete task (enabled only when no flagged field is unresolved).
- **Data bindings.** `fields[*].value`, `.confidence`, `.evidence.page`, `.evidence.bbox`, `.review_required`, `.validation`, `flags[*]`.
- **Linked requirements.** FR-REV-001 to 003, FR-CLS-004, FR-CNF-002, GR-009, SEC-007.

SCR-03: source on the left, flagged fields first on the right



Selecting a field highlights its box on the page; Complete stays off until every flagged field is resolved.

SCR-03 wireframe · desktop, populated state with fictional data

The wireframe shows the populated state; the other states in the table change only the regions they name.

State	What the mock shows
Loading	Skeleton for viewer and panel; Complete disabled
Populated	Flagged fields on top, resolved count "3 of 5 resolved"
Low-confidence handwriting	Field shows chip "Low confidence 62%" and the cropped image next to the value
Validation failed	Inline message with the rule (for example "PAN format invalid"), Complete blocked
Arithmetic mismatch	Banner at top naming the fields that disagree, with the computed sum
Unknown type	Banner "Document type not recognised" with type selector, no field panel
Task claimed by someone else	Read-only with the owner's name
PII masked	Values masked with a reveal action only for roles allowed by SEC-007; each reveal is logged
Error	Page image failed to load; retry button; task stays claimed

8.5 Mock generation prompt template

Used with the screen spec as input to an AI design tool or a front-end generator.

```
Generate a high-fidelity mock for <SCR-ID> <name> of an on-premise document
extraction product.
Users: <persona>. Width: <1440|1024>. Theme: <light|dark>.
Use only the regions, components, states and actions in the attached screen
spec.
Render state: <state name>.
Use synthetic data only (fictional names, fake ID numbers). Show confidence
as text plus icon.
Annotate each interactive element with its requirement ID and data binding.
Do not add features, screens or copy that are not in the spec.
```

9. Guardrails

In lending and clinical settings an unverified guess is worse than a missing value. Guardrails therefore favour null and human review over completeness. They apply at four points: before the model (input), around the model (generation), after the model (output) and around the system (operations).

ID	Point	Guardrail	Mechanism	Acceptance test	Pri
GR-001	Input	Only supported, clean files enter the pipeline.	Magic-byte check, size and page limits, malware scan, decompression-bomb limit.	Fuzz corpus of malformed files: 0 crashes, 100% rejected with a code.	P0
GR-002	Output	No value without evidence.	A non-null value must carry page, bbox and a snippet that contains it after normalisation; otherwise it is set to null and flagged.	Seeded test: model returns a value with fake evidence; result is null and `EVIDENCE_MISSING`.	P0
GR-003	Generation	Null by default.	System prompt and schema instruct: "If not legibly visible, return null. Never guess." Absent and illegible are distinct reasons.	Blank-field benchmark: false-fill rate ⩽ 0.5% of blank fields.	P0
GR-004	Input	No forced classification.	Below-threshold or no matching schema yields `UNKNOWN` and review; extraction does not run.	Out-of-domain set: ⩾ 98% routed to `UNKNOWN`.	P0
GR-005	Generation	Output must conform to the schema.	Grammar-guided decoding against the compiled JSON Schema; no free text.	1,000 runs, 0 exceptions (M3).	P0
GR-006	Input	Document content is data, never instructions.	Document text is passed only inside delimited data fields; the prompt states that instructions inside documents are ignored; output is limited to schema fields so an injected instruction has no channel.	Injection test set ("ignore previous instructions…", hidden white text, instructions in stamps): 0 schema deviations, 0 behaviour changes.	P0

ID	Point	Guardrail	Mechanism	Acceptance test	Pri
GR-007	Generation	The model has no tools, no network and no memory across documents.	Inference runs in a network-isolated container; no function calling; context cleared per document.	Egress test from the inference container: all outbound attempts blocked and alerted.	P0
GR-008	Output	Validation gate before approval.	Format, checksum, arithmetic, chronology and cross-field rules (FR-VAL) run on every document; any failure blocks auto-approval.	Rule test suite 100% pass; seeded violations all blocked.	P0
GR-009	Output	Low confidence goes to a person.	Per-field thresholds; handwriting-derived fields default to 0.75; reviewer must resolve every flagged field.	Handwriting set: 0 auto-approved fields below threshold.	P0
GR-010	System	The engine never decides.	No endpoint or connector triggers underwriting, credit, claim or clinical action; output is data for downstream systems that own the decision.	Design review plus connector test: connectors write only extracted data to staging objects.	P0
GR-011	System	No silent change in behaviour.	Any change to a model, prompt, schema or rule version requires a full benchmark regression; release blocked if M1, M3 or M6 degrade beyond tolerance (F1 − 1 pt, calibration + 1 pt).	CI gate demonstrated by a deliberately degraded model being blocked.	P0
GR-012	System	Learning needs approval.	Reviewer corrections go to a candidate list; adding to the benchmark or a fine-tuning set needs a second person's approval and a PII-handling check.	Attempt to auto-ingest a correction fails.	P1
GR-013	System	Only approved providers run.	Provider policy flag lists allowed model endpoints; default is local only; an external endpoint fails closed unless a signed policy enables it.	Configure an external URL with default policy: startup refused.	P0

ID	Point	Guardrail	Mechanism	Acceptance test	Pri
GR-014	System	No PII in prompts to non-local services, logs or telemetry.	PII classes from the schema drive masking; log scrubbers; telemetry carries counts and codes only.	Log scan after the full test suite finds 0 PII values.	P0
GR-015	System	Kill switch.	Admin can pause auto-approval (all documents go to review) or the whole pipeline in under 1 minute, without a deploy.	Toggle test with timing; state change audited.	P0
GR-016	Output	Reproducibility.	Deterministic decoding (temperature 0, fixed seed), pinned model and prompt versions, stored with each run.	Re-run of 100 stored runs yields identical outputs.	P0
GR-017	Output	Honest uncertainty.	Reported confidence is calibrated; the UI shows confidence and null reasons plainly and never fills a gap with an estimate.	Calibration gap ⩽ 3 points (M6).	P0

9.1 Failure handling

ID	Requirement	Pri
GR-018	IF a stage times out or fails, THEN the system SHALL retry up to 3 times with back-off, then mark the document `failed` with a reason code and create a review task; no partial output SHALL be marked approved.	P0
GR-019	IF the model returns output that fails schema validation despite decoding constraints, THEN the system SHALL discard it, log a defect event and route the document to review.	P0
GR-020	IF GPU capacity is exhausted, THEN the system SHALL queue and report an estimated wait, not degrade to a smaller unvalidated model.	P1

10. Security, privacy and compliance

The controlling constraint is that PII and PHI never leave the customer's boundary. Security requirements below are testable controls; Section 11 maps them to NIST CSF 2.0.

10.1 Security requirements

ID	Requirement	Pri	Acceptance criteria
SEC-001	Users SHALL authenticate through the customer's identity provider (OIDC or SAML); administrators and schema approvers SHALL use MFA.	P0	Local passwords disabled by default; login without MFA as admin is refused.
SEC-002	API clients SHALL use OAuth 2.0 client credentials or scoped API keys; tokens expire within 1 hour (keys rotate at most every 90 days).	P0	Expired token returns `401`; scope test matrix passes.
SEC-003	Access SHALL be role-based and least-privilege with roles Analyst, Reviewer, Schema author, Schema approver, Integrator, Auditor and Admin; no user SHALL both author and approve the same schema version.	P0	Permission matrix test: every role × endpoint outcome matches the matrix.
SEC-004	All traffic SHALL use TLS 1.2 or higher, and service-to-service traffic SHALL use mutual TLS.	P0	Scan finds no plaintext listener; weak ciphers rejected.
SEC-005	Stored documents, extracted data, adapters and backups SHALL be encrypted at rest with AES-256, using keys held in the customer's KMS or HSM, separate per tenant.	P0	Storage inspection shows only ciphertext; revoking a tenant key makes its data unreadable.
SEC-006	Data, queues, caches and schemas SHALL be isolated per tenant or business unit; cross-tenant access SHALL be impossible through the API.	P0	Cross-tenant ID guessing test returns `404` for 100% of attempts.

ID	Requirement	Pri	Acceptance criteria
SEC-007	PII fields (schema `pii_class`) SHALL be masked by default in UI, API and exports; unmasking needs a role permission and SHALL be logged with a reason.	P0	Reviewer without permission sees masked values; each reveal creates an audit event.
SEC-008	Audit logs SHALL be append-only and tamper-evident (hash-chained), covering logins, uploads, views of unmasked data, corrections, schema changes, approvals, config changes and exports.	P0	Altering a stored event breaks chain verification, which raises an alert.
SEC-009	Every uploaded file SHALL be scanned for malware before any processing, and the scanner signatures SHALL be updated without internet access (offline update bundle).	P0	EICAR test blocked; signature age shown in SCR-09.
SEC-010	Retention SHALL be configurable per tenant and document type; originals and derived data SHALL be deleted by crypto-shred or secure delete on expiry or on request, with a deletion certificate event.	P0	Delete request leaves no recoverable file, text, embedding or log value; verified by storage scan.
SEC-011	The deployment SHALL deny all outbound network traffic by default and allow only named internal destinations.	P0	Network policy test: connections to the internet from every service fail.
SEC-012	Logs, metrics, traces and error messages SHALL NOT contain document content or PII.	P0	Automated scan of all telemetry after the test suite finds none.
SEC-013	Secrets SHALL be held in a secrets manager, never in images, code or config files, and SHALL be rotatable without downtime.	P0	Secret scan on repo and images finds none; rotation drill passes.

ID	Requirement	Pri	Acceptance criteria
SEC-014	Model weights, containers and dependencies SHALL be pinned by hash, signature-verified at deploy, and listed in an SBOM; model licences SHALL be reviewed for commercial and on-premise use.	P0	Deploy with a modified weight file is refused; SBOM generated per release.
SEC-015	The product SHALL follow a secure SDLC: threat model per release, SAST, dependency and container scanning in CI, and an independent penetration test before general availability.	P0	No open critical or high findings at release; pen-test report on file.
SEC-016	Backups SHALL run on a schedule with tested restore; RPO ⩽ 15 minutes for metadata and RTO ⩽ 4 hours (to confirm).	P0	Quarterly restore drill meets RPO and RTO.
SEC-017	The API SHALL enforce rate limits and request size limits per client.	P0	Load test beyond limit returns `429`, service stays within latency SLO for other tenants.
SEC-018	Fine-tuning data and adapters SHALL stay in the customer's boundary, be access-controlled and lineage-tracked (source, consent, approver), and be deletable.	P1	Lineage report lists every record in a training set; deleting a source removes it from future runs.
SEC-019	Data residency SHALL be configurable so that storage and processing stay in a named location.	P0	Deployment manifest pins location; no component can be placed outside it.
SEC-020	UI sessions SHALL time out after inactivity (default 15 minutes) and the UI SHALL set standard protections (CSRF, CSP, secure cookies).	P0	Browser security-header scan passes.
SEC-021	The system SHALL raise alerts for repeated failed logins, unusual export volume, unmasking spikes, egress attempts, audit-chain failures and model-file changes.	P0	Each condition triggers an alert in a simulation.

ID	Requirement	Pri	Acceptance criteria
SEC-022	An incident response plan and runbooks (data exposure, model compromise, ransomware, bad model release) SHALL exist and be exercised at least annually.	P0	Tabletop exercise report on file.
SEC-023	Critical vulnerabilities in components SHALL be patched or mitigated within 7 days, high within 30 days.	P1	Patch-age report in SCR-10.

10.2 Privacy and regulatory alignment

The product provides controls; compliance is a shared responsibility with the customer. Obligations apply by deployment.

Regime	What the product must support	Related requirements
India DPDP Act 2023	Purpose limitation, retention limits, deletion on request, breach logging, data processed in India	SEC-010, SEC-019, SEC-008
HIPAA (where PHI of US persons is processed)	Access control, audit controls, integrity, transmission security, minimum necessary	SEC-003, SEC-004, SEC-007, SEC-008
GDPR (where applicable)	Data minimisation, erasure, records of processing, automated-processing transparency	SEC-010, FR-AUD-001, GR-010
RBI and IRDAI expectations for regulated entities	Data localisation, audit trails, vendor risk evidence	SEC-011, SEC-019, SEC-014
ISO/IEC 27001 and SOC 2	Control evidence produced by the product for the customer's audit	Section 11

Legal review of the exact obligations per customer is a dependency (Section 15); this table is a requirement-mapping aid, not legal advice.

11. NIST CSF 2.0 alignment

The product is designed so a customer can show NIST Cybersecurity Framework 2.0 coverage across all six Functions. The table maps each CSF 2.0 Category to the product requirements that implement it and to the evidence the product produces. Controls that depend on the customer's own organisation (for example governance policy) are marked **Customer** and the product supplies the evidence hooks.

Function	Category	Product requirements	Owner	Evidence produced
Govern (GV)	GV.OC Organizational Context	Section 2.3 non-goals; GR-010 (engine never decides); Section 10.2 regime map	Product	PRD baseline, regulatory mapping per customer
	GV.RM Risk Management Strategy	Section 15 risk register; GR-011 change gate; thresholds in schemas	Product, Customer	Risk register, threshold configuration history
	GV.RR Roles, Responsibilities, Authorities	SEC-003 roles and separation of duties; SCH-005 four-eyes approval	Product, Customer	Role matrix export from SCR-09
	GV.PO Policy	SEC-010 retention; SEC-011 egress; GR-013 provider policy as enforced settings	Customer	Policy settings with change history
	GV.OV Oversight	SCR-10 dashboard; metrics M1 to M10; SEC-008 audit	Product, Customer	Monthly accuracy, STP and incident report
	GV.SC Supply Chain Risk Management	SEC-014 pinned and signed models, SBOM, licence review; SEC-015 dependency scanning	Product	SBOM, signature verification log, model licence register

Function	Category	Product requirements	Owner	Evidence produced
Identify (ID)	ID.AM Asset Management	Inventory of models, schemas, prompts, services and data stores; SEC-014 SBOM	Product	Asset inventory view in SCR-10
	ID.RA Risk Assessment	SEC-015 threat model per release; GR-006 injection tests; Section 15	Product	Threat model, pen-test report, vulnerability reports
	ID.IM Improvement	FR-REV-004 and GR-012 controlled feedback loop; SEC-022 exercise findings; GR-011 regression results	Product	Lessons-learned log, benchmark growth record
Protect (PR)	PR.AA Identity Management, Authentication, Access Control	SEC-001, SEC-002, SEC-003, SEC-006, SEC-007, SEC-020	Product	Access reviews, permission test results
	PR.AT Awareness and Training	Reviewer and admin onboarding guide; in-product masked-data notices	Customer, Product	Training completion (customer), guide versions
	PR.DS Data Security	SEC-004, SEC-005, SEC-007, SEC-010, SEC-012, SEC-018, SEC-019, GR-014	Product	Encryption configuration, deletion certificates, telemetry scan
	PR.PS Platform Security	SEC-009, SEC-013, SEC-014, SEC-015, SEC-023; GR-001, GR-007 hardened, isolated inference	Product	Hardening baseline, scan results, patch-age report

Function	Category	Product requirements	Owner	Evidence produced
	PR.IR Technology Infrastructure Resilience	SEC-011 network policy; SEC-016 backup and restore; NFR-004 availability; NFR-005 horizontal scaling; GR-020	Product, Customer	Restore drill report, capacity test
Detect (DE)	DE.CM Continuous Monitoring	SEC-021 alerts; SEC-008 chain verification; NFR-010 observability; GR-007 egress alerts	Product	Alert catalogue, monitoring dashboards
	DE.AE Adverse Event Analysis	Audit explorer SCR-08; drift and anomaly detection on confidence and error patterns (NFR-011)	Product	Correlated event timeline, drift report
Respond (RS)	RS.MA Incident Management	SEC-022 runbooks; GR-015 kill switch; GR-018 failure handling	Product, Customer	Runbooks, kill-switch audit events
	RS.AN Incident Analysis	FR-AUD-001 reproducible runs; SEC-008 tamper-evident logs; SCR-08 run reproduction	Product	Reproduction package per incident
	RS.CO Incident Response Reporting and Communication	Notification templates and contact configuration; breach-relevant audit export	Customer, Product	Exportable incident timeline

Function	Category	Product requirements	Owner	Evidence produced
	RS.MI Incident Mitigation	GR-015 pause auto-approval; SEC-005 key revocation; SEC-013 secret rotation; model rollback (GR-011)	Product	Mitigation actions in audit log
Recover (RC)	RC.RP Incident Recovery Plan Execution	SEC-016 backup and restore; versioned schemas and models with rollback (SCH-006)	Product, Customer	Recovery drill report
	RC.CO Incident Recovery Communication	Status banner and recovery notices in the UI; post-incident report template	Customer, Product	Post-incident review record

11.1 CSF-specific requirements

ID	Requirement	Pri	Acceptance criteria
SEC-024	The product SHALL ship a CSF 2.0 profile (current and target) in machine-readable form listing, per Category, the implementing requirements and test evidence.	P0	Profile file validates; every Category in the table above has at least one requirement and one test.
SEC-025	The product SHALL produce a compliance evidence bundle (configuration, access report, audit extract, vulnerability and patch status, restore drill result) on demand from SCR-09.	P1	Bundle generated in under 10 minutes and contains no document content.

ID	Requirement	Pri	Acceptance criteria
SEC-026	The CSF profile SHALL be reviewed each release; a requirement change that affects a Category SHALL update the profile in the same change.	P0	CI check fails when a requirement tagged with a CSF Category changes without a profile change.

The CSF covers cybersecurity outcomes. AI-specific risks (hallucination, drift, bias in handwriting recognition) are handled by Section 9 and are consistent with the NIST AI Risk Management Framework, which the customer may use alongside CSF.

12. Non-functional requirements

Reference hardware for performance targets: a single server with one 24 GB GPU (RTX 3090 or 4090 class), 16 cores and 64 GB RAM. Confirming the customer's actual hardware is an open item (Section 15).

ID	Area	Requirement	Target	Pri
NFR-001	Latency	Upload to validated JSON, per page, on the reference GPU.	p95 ⩽ 3.0 s	P0
NFR-002	Latency	Same on CPU-only fallback with the same models.	p95 ⩽ 10 s	P1
NFR-003	Throughput	Sustained throughput on the reference server.	⩾ 15 pages/min (to validate)	P0
NFR-004	Availability	Service availability for a single-site deployment; high-availability option documented.	99.5% monthly (HA: 99.9%)	P0
NFR-005	Scalability	Add worker nodes to scale throughput linearly up to 4 nodes; queue absorbs month-end bursts.	⩾ 85% scaling efficiency	P1
NFR-006	Reliability	Retries and idempotency on all mutating calls; at-least-once processing with no lost documents on node failure.	0 lost documents in failure-injection test	P0

ID	Area	Requirement	Target	Pri
NFR-007	Model footprint	Parser stage VRAM; extractor stage VRAM with quantisation.	Parser < 2 GB; extractor within 8 to 16 GB	P0
NFR-008	Model flexibility	Parser and extractor sit behind interfaces; swapping a model needs configuration and a regression run, not code changes in other stages.	Swap demonstrated for one model per stage	P0
NFR-009	Deployment	One container build deploys to on-premise, private cloud or VPC with Helm or Compose; offline installation bundle.	Install with no internet in under 1 day	P0
NFR-010	Observability	Dashboards for accuracy, STP, confidence distribution, error patterns, latency, GPU use and cost per page; no PII.	All listed panels live	P1
NFR-011	Drift	Weekly comparison of confidence and error distributions against baseline; alert on shift beyond threshold.	Alert within 24 hours of a seeded shift	P1
NFR-012	Usability	Reviewer completes a typical flagged document (5 flagged fields) quickly.	Median ⩽ 90 s (to validate)	P1
NFR-013	Accessibility	UI meets WCAG 2.2 AA.	Automated and manual audit pass	P0
NFR-014	Localisation	UI text in English at MVP; Indic script rendering in data views.	Rendering check per script	P0
NFR-015	Fine-tuning	PEFT/QLoRA fine-tuning of the extractor runs locally on the reference GPU; adapters are versioned and revertible.	One adapter trained and rolled back in test	P1

ID	Area	Requirement	Target	Pri
NFR-016	Cost	Cost per page (compute plus licences) reported per tenant; per-page pricing needs a metering record per processed page.	Metering matches processed page count within 0.1%	P0
NFR-017	Maintainability	Minimum 80% unit-test coverage on validation, normalisation and routing code; every pipeline stage has a contract test.	Coverage gate in CI	P0

13. Evaluation and test strategy

The benchmark decides when a phase is done, so it is built before the model work and is never used for tuning.

13.1 Benchmark set

- **Splits per document type:** development (used freely), tuning (used for prompt and threshold selection) and a locked test set that is run only at phase gates and after any model, prompt or schema change.
- **Size:** the approach document asks for 5–10 anonymised scans per type to start. That is enough for development only. Gate-grade size per MVP type is set with the customer; the PRD proposes at least 100 documents per type in the locked set (to confirm), so that a 1-point change in F1 is distinguishable.
- **Strata:** clean print, skewed or low-resolution scans, stamps or signatures over text, character grids, dense checkbox grids, handwriting, each Indic language, and multi-hospital layouts for DOC-03.
- **Ground truth:** double-keyed by two annotators with adjudication, stored with the document hash and annotator IDs; PII-protected like production data.
- **Synthetic supplement:** MOCK-005 samples extend coverage but never count towards a gate on their own.

13.2 Test layers

Layer	What it proves	Run when	Blocks release
Unit	Validators, normalisers, rule engine, routing, confidence maths	Every commit	Yes
Contract	OpenAPI and event schemas; each pipeline stage's input and output contract	Every commit	Yes
Pipeline integration	Upload to envelope on synthetic documents	Every merge	Yes
Benchmark regression	M1 to M6 on dev set per merge; locked set at gates and model changes	See left	Yes (GR-011)
Adversarial	Prompt-injection set (GR-006), malformed files (GR-001), blank-field false-fill (GR-003)	Every release	Yes
Security	SAST, dependency and image scans, egress test, cross-tenant test, pen test	CI; pen test before GA	Yes
Performance	NFR-001 to NFR-005 on reference hardware, burst test	Every release candidate	Yes
Resilience	Failure injection, backup restore, kill switch	Quarterly and before GA	Yes
UX conformance	Built screens match signed-off mocks and states (MOCK-003, 009); accessibility audit	Per screen	Yes

13.3 Measurement rules

- **Field match:** exact match after normalisation for identifiers, dates, amounts and enums; token-level F1 for free text. The rule per field type is fixed before the first run.
- **Null handling:** predicting a value where ground truth is null counts as a false positive and is reported separately as the false-fill rate.
- **Calibration:** confidence is binned in 10 equal bins; the gap is the absolute difference between mean confidence and accuracy, weighted by bin size.

- **STP:** share of documents auto-approved whose approved values match ground truth; auto-approved documents with any wrong key field are counted as escapes and reported as a separate number that SHALL be 0 on the locked set for DOC-01 to DOC-03 at the MVP gate (to confirm with the customer).

13.4 Definition of done

A requirement is done only when all of these hold:

- [ ] Code merged with tests that cite the requirement ID.
- [ ] Acceptance criteria in the PRD pass in CI.
- [ ] Guardrail and security tests tagged to the requirement pass.
- [ ] Telemetry and audit events exist and contain no PII.
- [ ] The API contract or screen mock is updated and signed off where affected.
- [ ] The CSF profile (SEC-024) is updated where a Category is affected.
- [ ] Documentation and runbook entry updated.

14. Roadmap and phase gates

Phases are gated by evidence, not dates. Durations are set after capacity planning and hardware confirmation. The approach document's advice is followed: stabilise the pipeline on the two structural extremes (CASA form and discharge summary), then treat the rest as schema work.

MVP — prove the pipeline on one document per domain.

- Types: DOC-01 CASA, DOC-02 Insurance application, DOC-03 Discharge summary.
- Capabilities: printed text and clean-to-skewed scans, classification, schema-guided extraction with null handling, evidence and confidence, core validations, review workbench, JSON API, audit, security baseline, English handwriting flagged for review only.
- Screens: SCR-01 to SCR-09 mocks signed off and built.

v1 — widen coverage and connect.

- Types: DOC-04 to DOC-07, plus DOC-08 brochures and policy wordings.
- Capabilities: handwriting reading, Hindi, Telugu, Tamil, Malayalam (each language ships only when it passes), bundle splitting, schema library, CSV/XML, first system connector, operations dashboard SCR-10, evidence bundle (SEC-025).

v2 — reason across documents and improve safely.

- Capabilities: natural-language schema drafting, case-level cross-document reasoning, agentic re-read and missing-document requests, approved-feedback fine-tuning loop, high-availability reference deployment, wider connector catalogue.

Gate	Entry to	Criteria (all on the locked test set unless stated)
G0	Build	PRD signed off; decisions D1 to D5 confirmed; sample documents and hardware confirmed; benchmark process agreed
G1	MVP release	M1 ⩾ 92% all fields and ⩾ 95% key fields on DOC-01 to DOC-03; M3 = 100%; M4 ⩾ 98%; M5 ⩾ 60% with 0 escapes; M6 ⩽ 3 points; M7 met; M9 = 0; all P0 guardrails and security tests pass; pen test has no open high findings; CSF profile complete; mocks signed off
G2	v1 release	G1 criteria hold for all seven types; M5 ⩾ 80% on structured types; per-language CER meets M2 for each shipped language; connector end-to-end test passes; SEC P1 items pass
G3	v2 release	Case-level checks measured on a case benchmark; fine-tuning loop passes lineage and approval tests (SEC-018, GR-012); HA drill passes

15. Risks, assumptions, dependencies and open questions

15.1 Risks

ID	Risk	Impact	Mitigation	Related
R1	Published model accuracy (for example >95% on OmniDocBench) does not carry over to Indian bank and hospital forms.	Targets missed	Treat vendor figures as hypotheses; measure on the locked set at G0 spike; keep parser behind an interface.	NFR-008, M1

ID	Risk	Impact	Mitigation	Related
R2	Handwritten prescriptions have high error rates and ambiguous drug names.	Clinical risk	Review below 0.75, show crop beside value, never auto-approve; set STP expectations accordingly.	GR-009, FR-CNF-002
R3	A 7B local model hallucinates or mis-assigns values in dense forms.	Wrong data in downstream systems	Null by default, evidence required, grammar-guided decoding, validation gate, second pass.	GR-002, GR-003, GR-005
R4	Customer GPU is smaller than assumed, so latency targets fail.	SLA breach	Confirm hardware at G0; quantisation options; CPU fallback target; queue with wait estimate.	NFR-001, NFR-002
R5	Malayalam, Telugu or Tamil accuracy is below the bar with the chosen models.	Launch languages slip	Per-language gate; ship languages independently; evaluate alternative parsers.	FR-PRS-007
R6	Prompt injection through document content.	Corrupted output	Data/instruction separation, schema-only outputs, no tools, adversarial test set.	GR-006, GR-007
R7	Sample documents or ground truth arrive late or are too few.	Cannot gate	Synthetic generator for development; contractual sample commitment; gate size set early.	MOCK-005, §13.1
R8	Model licence prevents commercial on-premise use.	Rework	Licence review at G0; alternatives behind the same interface.	SEC-014
R9	Reviewer fatigue causes accepting wrong values.	Escapes	Evidence highlight, no pre-checked accept-all, escape audit sampling of auto-approved documents.	FR-REV-001

ID	Risk	Impact	Mitigation	Related
R10	Scope creep to "any document".	Missed commitments	Publish supported types with measured accuracy; new type only through schema and gate.	§3.1

15.2 Assumptions and dependencies

- The customer provides representative anonymised samples and ground truth for each launch type.
- The customer provides GPU servers, an identity provider, a KMS or HSM, and the network policy to deny egress.
- Access to policy-administration, loan-origination and claims systems (sandbox) is available for connector work in v1.
- Models named in the approach document (PaddleOCR-VL-1.6 or GLM-OCR for parsing; quantised Qwen2.5-7B-Instruct for extraction, served by Ollama or vLLM) are the starting candidates, not commitments; the PRD binds interfaces and measured outcomes.
- Legal review of DPDP, HIPAA, GDPR, RBI and IRDAI obligations is done per customer outside this PRD.

15.3 Answers carried over from the outline

Question	Answer recorded	How this PRD treats it
Anchor industries	InsurTech, FinTech, Healthcare	MVP types span all three (D5)
Deployment	SaaS first	Conflicts with on-premise constraint; resolved by D1, needs sign-off
Accuracy and STP target	"80% above"	Read as STP of 80% or more; set as the v1 target, with 60% at MVP (M5); confirm
Pricing	Per page	NFR-016 metering requirement
Languages at launch	English, Hindi, Telugu, Tamil, Malayalam	Per-language gate (FR-PRS-007)

15.4 Open questions and decisions needed

- [ ] **D1 to D5 sign-off.** Confirm the five decisions in Section 1, especially on-premise-first versus SaaS-first.
- [ ] **Hardware.** What GPU model, VRAM and CUDA availability will customers run? (approach checklist)
- [ ] **Samples.** Can the customer supply 5–10 anonymised scans per type now, and a locked set of about 100 per MVP type by G1?
- [ ] **Existing schemas.** Does an organisational schema already exist for each document type? (approach checklist)
- [ ] **STP definition.** Is 80% an STP target or an accuracy target, and on which document types?
- [ ] **Escapes.** Is "0 wrong key fields among auto-approved documents" an acceptable MVP gate, or what rate is?
- [ ] **Operational targets.** Confirm availability (99.5%), RPO (15 minutes), RTO (4 hours) and throughput (15 pages/min) proposed here.
- [ ] **Identity.** Which identity providers must be supported at launch?
- [ ] **Handwriting languages.** Is English-only handwriting at MVP acceptable?
- [ ] **Model licences.** Who completes the licence and export review for the candidate models?
- [ ] **Healthcare data rules.** Which regime applies to the first healthcare customer (HIPAA, DPDP, or both)?

16. Spec-driven development handoff

The build follows the spec: requirements and mocks come first, tests are written from acceptance criteria, and code is accepted only when those tests pass. This section turns the PRD into work an engineer or coding agent can pick up without interpretation.

16.1 Specification repository layout

Proposed for the `icxbee-document-extractor` repository.

```
specs/
  prd/                # this PRD split by section, one file per section, IDs
unchanged
  schemas/            # document-type schemas (YAML) and the envelope JSON
Schema
  api/openapi.yaml    # generated and checked against API-001 to API-014
  screens/            # one spec per SCR-nn plus generated mocks and sign-
offs
  guardrails/         # GR tests: injection set, malformed files, blank-
field set
  security/           # threat model, CSF profile (SEC-024), SBOM config
  traceability.csv    # requirement ID, component, test ID, screen, CSF
category, status
tests/
  unit/ contract/ integration/ benchmark/ adversarial/ security/ performance/
```

16.2 Traceability matrix (by requirement family)

Requirement family	Component	Primary tests	Screens	CSF Category
FR-ING, GR-001	Ingestion and preprocessing service	Unit, fuzz, integration	SCR-01, SCR-02	PR.PS
FR-CLS	Classifier and schema binder	Benchmark (M4), out-of-domain set	SCR-03	PR.PS
FR-PRS	Visual parser service (local model)	Benchmark (M2), checkbox set	SCR-03	ID.AM, GV.SC
FR-EXT, GR-002 to 007	Extraction service, prompt and schema compiler	Benchmark (M1, M3), blank-field, injection	SCR-03, SCR-06	PR.DS, ID.RA
FR-VAL, GR-008	Validation and rules engine	Unit rule suite	SCR-03, SCR-05	PR.DS
FR-CNF, FR-REV	Confidence, routing, review service	Calibration (M6), UI conformance	SCR-03, SCR-04	GV.OV
SCH, API	Schema service, API gateway	Contract (API-014), lifecycle tests	SCR-05, SCR-07	PR.AA
FR-AUD, SEC-008	Audit service	Tamper test, reproduction test	SCR-08	DE.AE, RS.AN

Requirement family	Component	Primary tests	Screens	CSF Category
SEC-001 to 007	Identity, authorisation, masking	Permission matrix, cross-tenant	SCR-09	PR.AA, PR.DS
SEC-009 to 023	Platform hardening, backup, monitoring	Egress, restore drill, pen test	SCR-09, SCR-10	PR.PS, PR.IR, DE.CM
NFR	All	Performance, resilience	SCR-10	PR.IR
MOCK, SCR	UI	Mock conformance, accessibility	All	PR.AT

16.3 Build order (vertical slices)

Each slice ends with something demonstrable and passing its tests. Slices 1 to 6 use DOC-01 and DOC-03 only, as the approach document recommends.

Slice	Outcome	Main requirements
0	Repo, CI, spec lint, traceability check, synthetic document generator, benchmark harness skeleton, signed-off mocks for SCR-01 to SCR-04	MOCK-001 to 009, SEC-015, §13
1	Upload to preprocessed pages with idempotency and file safety	FR-ING-001 to 006, GR-001
2	Layout parsing to Markdown, tables, checkboxes, with bounding boxes	FR-PRS-001 to 005
3	Classification, schema binding, schema service with lifecycle	FR-CLS, SCH-001 to 007
4	Guided extraction with evidence, null handling and guardrails; first benchmark numbers	FR-EXT-001 to 007, GR-002 to 007
5	Validation, normalisation, confidence, routing and envelope	FR-VAL, FR-CNF, SCH-011 to 014
6	API, webhooks, audit and reproducibility	API-001 to 014, FR-AUD-001
7	Review queue and workbench, results and export screens	SCR-03, 04, 06; FR-REV

Slice	Outcome	Main requirements
8	Schema studio, intake and list screens; DOC-02 added	SCR-01, 02, 05; SCH-003, 008
9	Security hardening, masking, retention, monitoring, CSF profile and evidence	SEC-001 to 026, SCR-08, 09
10	Performance and resilience tests, locked-set run, gate G1 review	NFR, §13, G1

16.4 Task slice template (for engineers and coding agents)

```
Task: <slice>-<nn> <title>
Implements: <requirement IDs>
Spec files to read: <paths>
Inputs / outputs: <contracts, with schema paths>
Tests to write first: <acceptance criteria turned into tests, with IDs>
Guardrails and security checks: <GR and SEC IDs that apply>
Screens affected: <SCR IDs and the signed-off mock version>
Out of scope: <what must not be changed>
Done when: Definition of done in section 13.4 is met
```

16.5 Working rules for spec-driven builds

1. The PRD and the files under `specs/` are the source of truth. If code and spec disagree, the spec is changed first, with review, then the code.
2. Tests cite requirement IDs; the traceability check fails the build when a P0 requirement has no test.
3. A person or agent never invents behaviour that is not in the spec; an ambiguity becomes a comment on the PRD and a blocked task.
4. No real customer data in the repository, prompts, mocks, logs or tests (MOCK-004, SEC-012).
5. Each slice is reviewed against the guardrail and security checks before merge, not at the end.